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Research on non-invasive load identification method based on VMD

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Abstract

The non-intrusive load identification method can judge the type and working situation of the electrical load without interfering with the user's electricity consumption, which is an important part of smart grid technology. In this paper, a load identification method based on variational modal decomposition algorithm (VMD) is proposed, which can effectively identify different types of electrical loads. The effectiveness of VMD algorithm in the field of load identification is verified by conducting load recognition experiments. After optimizing the algorithm parameters, the Feature library membership threshold was tested and debugged, and finally a good load identification effect was obtained, which can be used as a means of identifying electricity load in urban management.

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Keywords: Load identification; VMD; Feature library

1. Introduction

With the economic development and energy consumption, the increase of electricity demand makes the power system increasingly complex, and the power supply presents complex and diversified [1]. Some electrical loads, such as E-Bikes, may affect grid stability or cause fires. Load identification technology can monitor the use of electrical appliances in time to actively respond to energy saving policies [2], and promote rational use of electricity by users [3,4]. Meanwhile, the power sector can get the consumption composition of residents in detail and provide data support for power sector [5,6]. Load identification can be mainly divided into two types: invasive load identification and non-invasive load identification [7]. Although the accuracy of invasive load identification is higher, but the cost is larger and the maintenance is difficult. Non-invasive load identification costs lower and easy installation. Non-invasive load identification has become the focus of research in recent years. Non-invasive load identification has been proposed for many years, and there are various identification methods. Load identification has always been the focus of research in the case of multiple electrical loads acting together. Many different types of loads exist in the power grid at the same time, and the waveform aliasing increases the difficulty of identification. In

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2011, literature [8] selected active power, reactive power and instantaneous transient energy of equipment startup as load characteristics for industrial and commercial loads through wavelet transform and Fourier transform, and finally combined with neural network algorithm for load identification. As a result, the identification accuracy is not high when multiple loads are started at the same time. Literature [9] proposed a load classification method based on FCM clustering method and combined it with VMD algorithm to improve the accuracy of load classification. Literature [10] proposed a method to identify the load of household appliances based on the closeness of transient power characteristics of household appliances switching. The active power and reactive power waveform vectors were selected as load characteristics, and the power transformation values before and after transient switching were selected as auxiliary load characteristics. The membership between the signal and the standard library was calculated, and the load was identified by the principle of maximum membership. If this idea is combined with advanced signal processing methods, it may better solve the problem of load identification.

VMD (Variational Modal Decomposition), which was first proposed by Konstantin [11], is a new adaptive signal processing method based on Wiener filter. The algorithm adopts a non-recursive processing strategy to determine the frequency center and bandwidth of each component by iteratively searching the optimal solution of the variational model. The VMD algorithm can effectively avoid the problems of modal aliasing, over envelope, under envelope, endpoint effect and so on [12]. VMD algorithm is widely used in signal denoising, rolling bearing fault recognition and other fields. VMD algorithm can effectively distinguish different fault characteristics of rolling bearings, which cause vibration waveform changes and aliasing. The waveform aliasing caused by electrical load is similar to that caused by rolling bearing fault. Whether VMD algorithm can also be used in specific electrical load identification field. Based on VMD algorithm, this paper proposes a method for specific electrical load identification by evaluating the correlation between load signal modal components and load characteristics to be identified, and carries out experimental verification.

2. VMD algorithm

VMD algorithm has the ability to decompose any signal into a series of band limited intrinsic mode functions [13]. VMD decomposition is mainly based on the concepts of Wiener filter, Hilbert transform and frequency mixing. In order to minimize the sum of estimated bandwidth, the original signal is introduced into the variational model for the optimal solution, and obtain the optimal decomposition of the signal.

- Construction of variational model

The original signal $s(t)$ consists of k component signals. The preset mode number is determined by Fourier transform and other methods. VMD assumes that the original signal $s(t)$ can be decomposed into k IMF components, and defines each IMF component as FM and AM signal.

$$U_k(t) = A_k(t)cos(\varphi_k(t)) \tag{1}$$

The analytic signal is obtained by Hilbert transform of $u_k(t)$:

$$\left(\delta\left(t\right) + \frac{j}{\pi t}\right) u_k\left(t\right) \tag{2}$$

In (2), $\delta(t)$ is the Dirac function; j is an imaginary unit.

Based on the frequency shift characteristic of the analytic signal, a center frequency $e^{-j\omega_k t}$ is mixed, and the spectrum of $u_k(t)$ is moved to the corresponding base band.

$$\left[\left(\delta\left(t\right) + \frac{j}{\pi t}\right) u_k\left(t\right)\right] e^{-j\omega_k t} \tag{3}$$

The cumbersome square of the signal gradient after frequency shift is calculated to estimate the bandwidth of each modal component. In order to ensure the minimum sum of the bandwidth of each IMF component, the variable component of the established constraint is:

$$\left.\begin{array}{c} \min\limits_{\{uk\},\{\omega k\}} \left\{ \sum\limits_k \left\| \partial_t \left[\left(\delta\left(t\right) + \frac{j}{\pi t}\right) u_k\left(t\right)\right] e^{-j\omega_k t} \right\|^2 \right\} \\ \sum\limits_k u_k = f \end{array}\right\} \tag{4}$$

In (4), f is the input signal; u_k is the kth modal component obtained after decomposition; ω_k represents the center frequency of the kth modal component, and $\delta(t)$ is the impulse function.

$$
\begin{aligned}
L\left(\{u_k\}, \{\omega_k\}, \lambda\right)\alpha = {} & \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) u_k(t) \right] e^{-j w_k t} \right\|_2^2 + \left\| f(t) - \sum_k u_{k(t)} \right\|_2^2 \\
& + \left\langle \lambda(t), f(t) - \sum_k u_{k(t)} \right\rangle
\end{aligned}
\tag{5}
$$

In (5), λ is the Lagrange multiplication operator; α is the secondary penalty factor; $f(t)$ is the signal to be decomposed; u_k is the kth variational modal component.

After the time–frequency domain transformation of the augmented Lagrange function, the corresponding extreme value solution can be obtained. Then the saddle point of the Lagrange function can be obtained through alternating direction multiplication and continuous iteration, and the optimal solution of Eq. (3) can be obtained, and finally the k modal components of $s(t)$ can be obtained.

3. Construction of the test system

In order to simulate the real electricity consumption, we built an experimental system using different types of real electrical equipment, including refrigerators, air conditioners, microwave ovens, desktop computers, electric kettles, etc., basically covering the electricity consumption. The electrical energy mainly affects the current waveform and power factor in the power grid, so the current waveform can be used to analyze the electrical energy characteristics.

3.1. Composition of the test system

The test system selects independent power supply for all test appliances to isolate interference from others. The test system consists of a sampling device and a current waveform comparison and identification program. When the electric appliance works, it will produce different current waveforms according to its own characteristics. The sampling device transmits the converted current waveform to the identification program. The identification program will compare the current waveform with the existing current characteristics in the feature library, so as to identify the type of electrical appliance. The structure diagram of the test system is shown in Fig. 1.

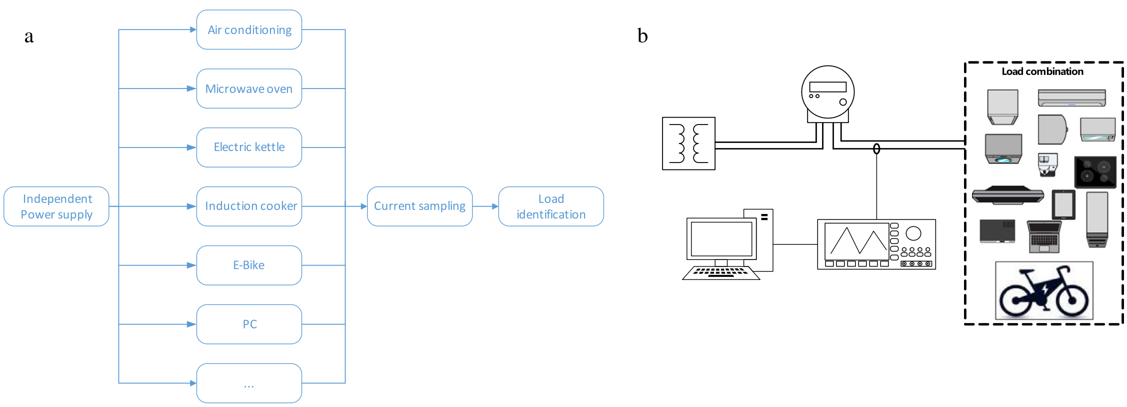

Fig. 1. Test system structure drawing.

3.2. Signal sampling and analysis program

The signal sampling and analysis program has the following functions: current waveform display, current waveform modal decomposition, sub-modal waveform display, sample rate adjustment, waveform data storage, and alarm function. The system program performs a variational modal decomposition of the acquired current waveform data and compares it with the reference mode comparison. If the system detects an electrical load that needs

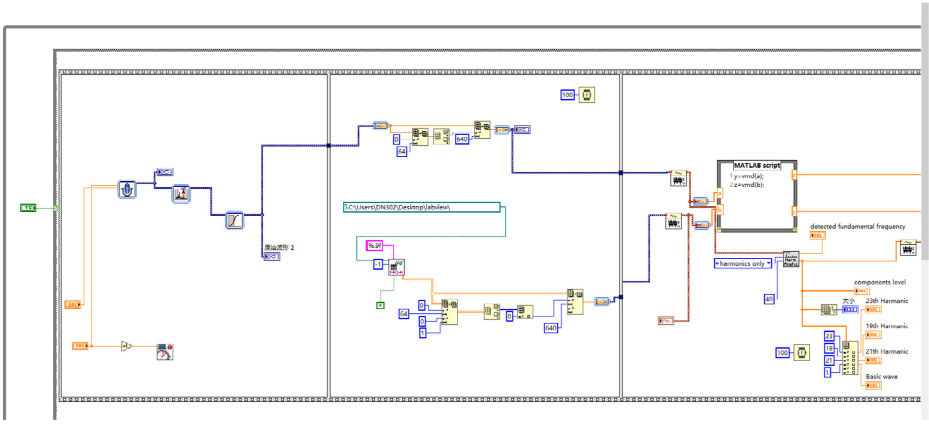

Fig. 2. Block diagram of program.

to be identified, it will be prompted through dialog boxes and Emails. The acquisition procedure is shown in Fig. 2.

We use current transformers and standard sample resistors to convert current signals into voltage signals that are easy to acquire. Uniformly collect the current waveform of 5 cycles of each electrical appliance, and normalized to ensure the consistency of the waveform data comparison. Normalization can effectively avoid the impact of amplitude changes on the algorithm's judgment. The VMD algorithm is used to decompose the waveform data, and the decomposed modal data are obtained as the reference modal standard data. The current waveform of the induction cooker, and E-Bike charger (hereinafter referred to as the charger) when working alone is shown in Figs. 3 to 4.

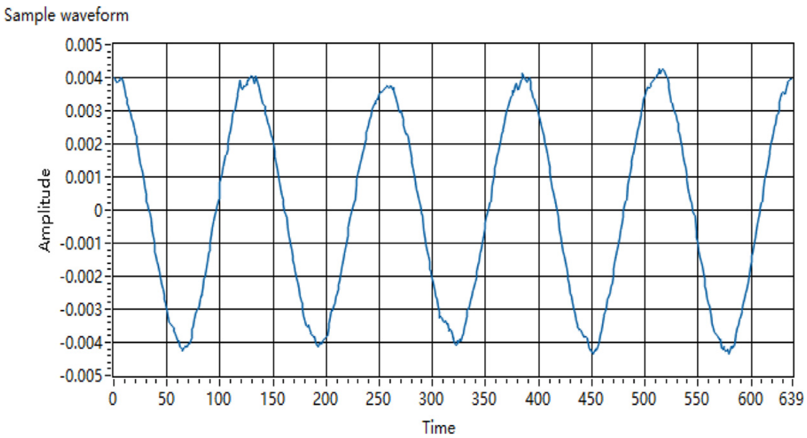

Fig. 3. Current waveform of induction cooker.

After the current waveform data is normalized, the VMD algorithm is used to decompose the signal into several decomposition modes.

The steps are following:

- Step 1: Initialize;
- Step 2: Execute the loop;
- Step 3: Update u_k and ω_k according to (6) and (7)

$$\hat{u}_k^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i \neq k} \hat{u}(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2} \tag{6}$$

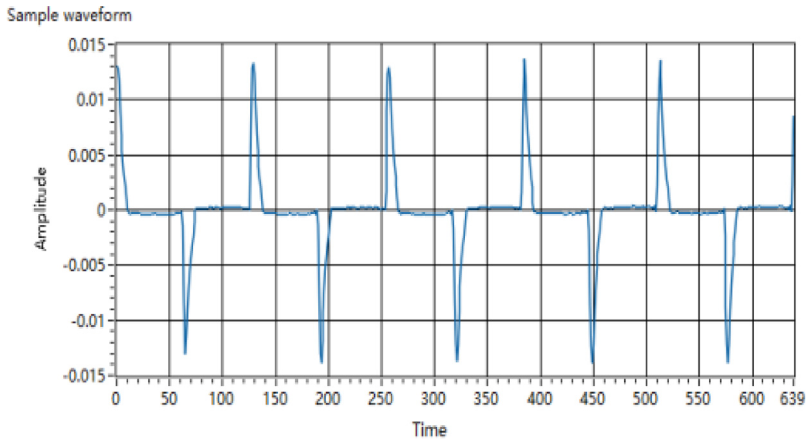

Fig. 4. Current waveform of E-Bike.

$$\omega_k^{n+1} = \frac{\int_0^\infty \omega \left|\hat{u}_k(\omega)\right|^2 d\omega}{\int_0^\infty \left|\hat{u}_k(\omega)\right|^2 d\omega} \tag{7}$$

- Step 4: Update λ according to (8);

$$\hat{\lambda}^{n+1}(\omega) \leftarrow \hat{\lambda}^{n}(\omega) + \tau\left[\hat{f}(\omega) - \sum_k \hat{u}_k^{n+1}(\omega)\right] \tag{8}$$

- Step 5: Repeat steps 2 to 4 until the iteration condition of (9) is satisfied.

$$\sum_k \left\|\hat{u}_k^{n+1} - \hat{u}_k^n\right\|_2^2 / \left\|\hat{u}_k^n\right\|_2^2 < \varepsilon \tag{9}$$

When a variety of electrical appliances work at the same time, the correlation between the modal data of the current waveform after VMD decomposition and each VMD reference modal data of the electrical appliance to be identified can be calculated, and a matrix composed of correlation coefficients can be obtained. If the elements in the correlation coefficient matrix exceed the set threshold, it indicates that there is a modal component in the current waveform with high correlation with the reference mode, thus proving that the electrical appliance to be identified is in working state.

3.3. Selection of test parameters

Through the pre sampling analysis of different electrical load conditions in the test system, it is found that the 3rd and 5th harmonics in the current waveform are the main components of harmonics, and the 20th to 40th harmonics account for very little. Considering the bandwidth limitation of data transmission and hardware cost of the sampling device in the future, on the premise of ensuring the effective sampling of current fundamental wave and main harmonic components, the sampling frequency is set to 6.4 kHz and the sampling bandwidth is set to 2 MHz, which can fully ensure the effective sampling of harmonic components within 40 times (see Fig. 5).

VMD algorithm mainly involves decomposing modal number k and penalty factor α, Noise tolerance (tau) and convergence criterion tolerance (tol). Penalty factors in these VMD algorithm association parameters α, Both the noise tolerance tau and the convergence criterion tolerance (tol) have corresponding empirical values, which can be directly adopted.

The decomposed mode number k is used to determine that the original signal is decomposed into several modal components for display. Too small or too large k value will cause the under decomposition and over decomposition of the original signal, affecting the judgment of the result. The comparison of sub modal center frequencies obtained by decomposing the current waveform of a single electrical appliance under different k values is shown in Table 1.

By comparison, it can be found that the center frequencies of mode 7 and mode 8 are similar when k is selected as 8, and there is aliasing, which indicates that the value of k should be selected as 7.

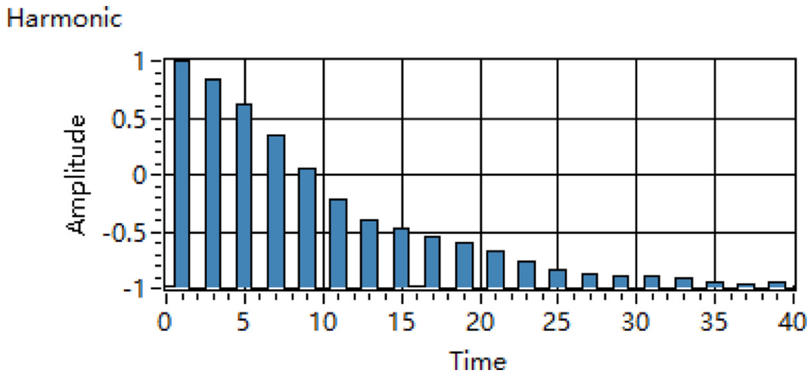

Fig. 5. Harmonic content.

Table 1. Comparison of modal center frequencies.

k	1	2	3	4	5	6	7	8
5	0.0120	0.0449	0.0888	0.1390	0.2097	/	/	/
6	0.0120	0.0443	0.0874	0.1352	0.1930	0.2780	/	/
7	0.0119	0.0436	0.0858	0.1321	0.1848	0.2477	0.3500	/
8	0.0099	0.0476	0.0850	0.1257	0.1728	0.2289	0.3306	0.0489

4. Test and analysis of results

4.1. Preliminary experiments

The test appliances are classified according to the working characteristics. The electric kettle and heater are pure resistive loads. Air conditionings and refrigerators are classified as compressor loads. Induction cooker and microwave oven are classified as inductive load. Notebook computers, routers, E-Bikes and other electrical appliances that use switching power supply fall into one category. Taking the charging behavior of E-Bike as an example. Turn on the electric bike charging switch and we can get the current waveform with the current sampling device (mentioned in Section 3). The current waveform of E-Bike charger working alone is shown in Fig. 6 (Sample waveform). The program performs VMD decomposition of the original current signal to obtain 7 decomposition modal signals, as shown in Fig. 6 (IMF1∼IMF7).

The different electrical appliances are combined to form a combination of electrical appliances such as E-Bike + microwave oven + induction cooker, and the current waveform at this time is collected, as shown in Fig. 7.

VMD decomposition was carried out for various electrical combinations, and correlation calculation was carried out by using reference modes of E-Bike chargers. Correlation matrix of all electrical combinations to E-Bike chargers could be obtained, as shown in Table 2.

Table 2. Correlation matrix data.

Correlation matrix data						
0.81880	0.13051	0.02501	0.00534	0.00314	0.00219	0.00181
0.03703	0.27837	0.56019	0.03076	0.01264	0.00714	0.00494
0.01721	0.05432	0.59818	0.36085	0.05539	0.02160	0.01183
0.00540	0.01671	0.06782	0.92087	0.03447	0.00777	0.00404
0.00295	0.00886	0.02244	0.00891	0.99587	0.00578	0.00290
0.00208	0.00573	0.01105	0.00290	0.00557	0.99502	0.01028
0.00142	0.00366	0.00582	0.00125	0.00123	0.00401	0.98972

By analyzing the matrix data, we found that if there are three data greater than 0.7 in the correlation matrix data, the E-Bike in the electrical combination can be judged. We set this condition as an experience value as the system

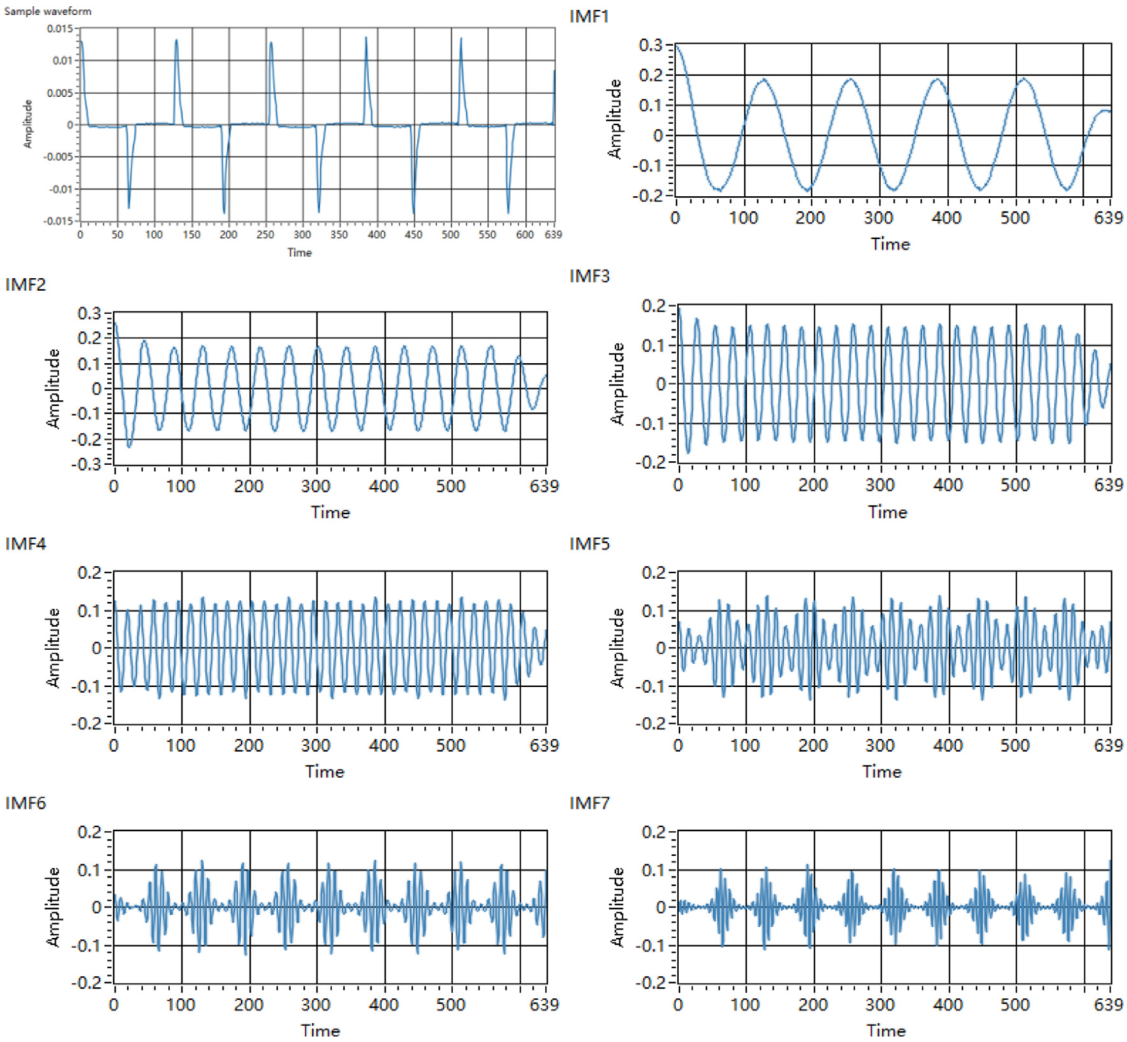

Fig. 6. Pre sampling analysis waveform diagram and 7 sub modal diagrams.

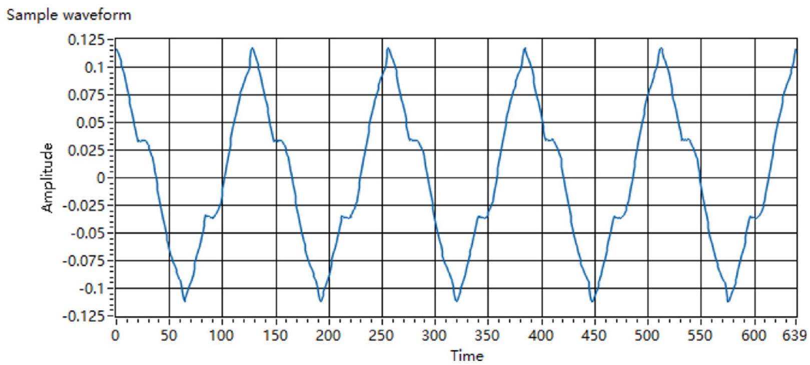

Fig. 7. Current waveforms when three appliances are operating at the same time.

criteria. Under this criterion, the detection effect of E-Bikes working together with different types of electrical appliances is shown in Table 3.

Table 3. Test results of E-Bike recognition.

	Number of experiments	Correct recognition times	Recognition rate
36 V E-Bike	200	167	83.5%
48 V E-Bike	200	163	81.5%
72 V E-Bike	200	166	83%

According to the test results, we found the system can effectively identify E-Bike from different combinations of appliances. The recognition rate is above 80%, indicating that it is feasible to use VMD algorithm for modal identification of current waveform.

Set the determination conditions to three correlation coefficients greater than 0.7 to determine the correlation for different types of loads. After many tests, the accuracy of the system to judge the electrical load can be obtained, as shown in Table 4.

Table 4. Accuracy of electrical load judgment.

	Number of experiments	Correct recognition times	Recognition rate
Air conditioner	200	168	84%
Electric kettle	200	172	86%
Microwave oven	200	167	83.5%

It can be seen that the identification rate can reach more than 80% only through the judgment basis of correlation coefficient, but the accuracy of this identification method does not reach the identification level of other load identification methods. Therefore, the identification method needs to be improved on the basis of VMD algorithm.

4.2. Improvements of the judgment mechanism and experiments

While optimizing the parameters of VMD algorithm, we improve the correlation data judgment mechanism according to the results of many experiments. Follow the formula (10), the correlation coefficient matrix is weighted to improve the weight ratio of specific data, so as to improve the recognition accuracy.

$$PA = \frac{(\frac{\tanh(A-A_0)}{\alpha} + 1)}{2} \tag{10}$$

In (10), A is the weighted sum of the correlation coefficient matrix, and A_0 is the threshold value of the weighted sum of the correlation coefficient matrix of the electrical load to be identified, α Take 0.2 as the parameter correction value, and the subsequent calculation will carry out iterative calculation according to the previous state to improve the identification accuracy. First, take an electric bicycle as the identification object. After testing different type of electrical equipment, the judgment effect of some electrical equipment when running alone is shown in Fig. 8.

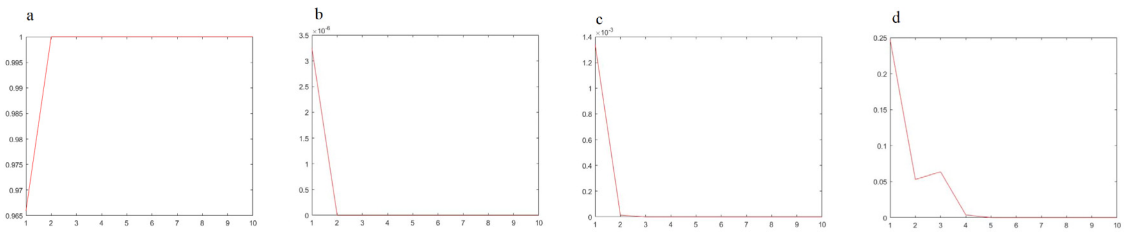

Fig. 8. (a) E-Bike; (b) electric kettle; (c) computer; (d) air conditioning.

This improved method can effectively improve the identification accuracy and speed of a single load. When the E-Bike is taken as the identification object, the sensitivity of the system to other types of loads is reduced, and the

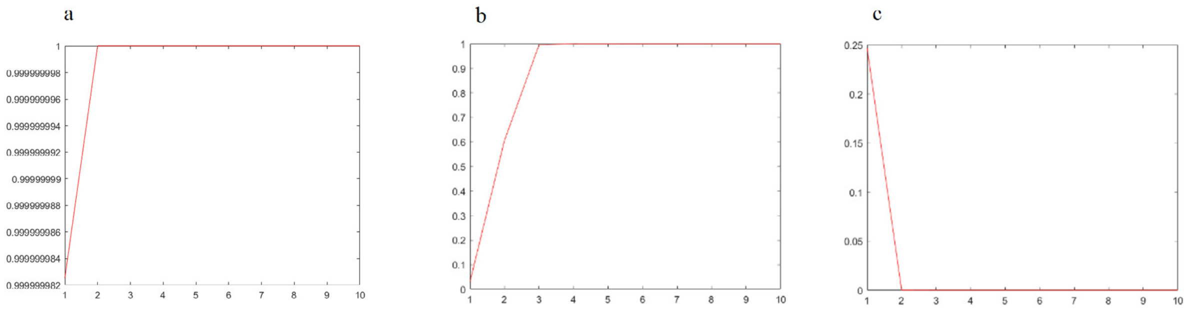

Fig. 9. (a) E-Bike + computer; (b) E-Bike + air conditioning; (c) computer + air conditioning.

false alarm rate is effectively reduced. The test results when the electric equipment is used in pairs are shown in Fig. 9.

The test results of more combinations of electric equipment are shown in Fig. 10 (see Table 5).

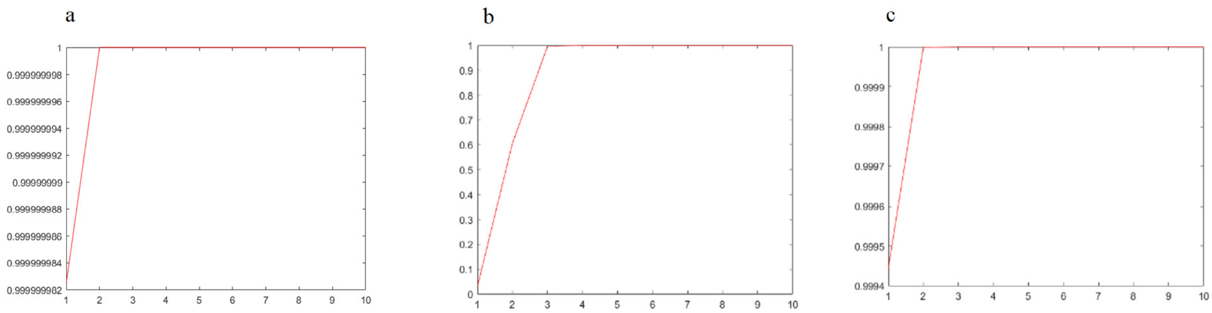

Fig. 10. (a) E-Bike + computer+ induction cooker; (b) E-Bike + air conditioning + computer + refrigerator; (c) 9 kinds of electric equipment.

Table 5. Statistics of experimental results.

	Number of experiments	Correct recognition times	Recognition rate
Single device	400	398	99.50%
Pairwise combination	400	391	97.75%
Complex combination	400	317	79.25%
Total	1200	1106	92.17%

It can be seen from the test that changing the system judgment threshold weight can improve the judgment accuracy of the original algorithm, especially for the case of less load types, the recognition rate is very high. After increasing the load type, the recognition rate of the system decreases. Overall, the existing load methods can effectively identify the specified load. In the future, in order to further improve the recognition rate of the system under complex conditions, we can improve the adaptability of the system and the accuracy of identification through the integration with adaptive algorithms such as machine learning.

5. Conclusions and prospects

Based on the VMD modal decomposition algorithm, this paper constructs an electrical load identification test system for E-Bike, and carries out the identification test. Through theoretical analysis and experiments, it is verified that the electrical load identification method based on VMD modal decomposition algorithm proposed in this paper is feasible, but how to further improve the electrical load identification rate needs further research and test, which can be carried out from two aspects: the optimization of VMD decomposition parameters and the enrichment of test categories. Due to the large volume of air conditioners, refrigerators and other electrical appliances used in the test system, it is not convenient to replace different brands of electrical appliances for comparative testing. In the future, we can form core equipment such as sampling devices and program modules into test devices that are easy to move, and carry out more extensive tests in the real residential electricity environment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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